

W01 - Command Line Environment on HPC

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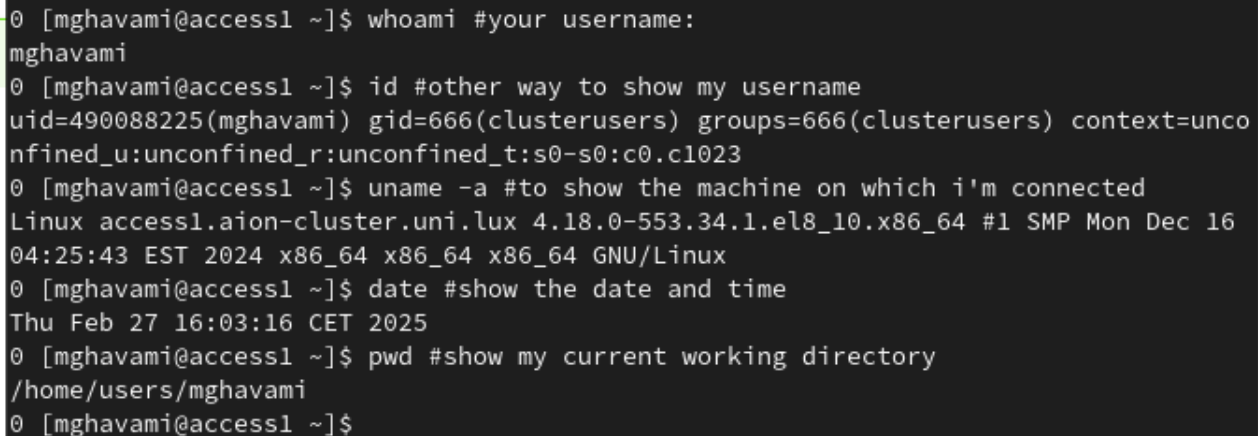
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1 Question 2:

Enter commands to show:

- your username
- the machine on which you're connected
- the date
- the current working directory



```
0 [mghavami@access1 ~]$ whoami #your username:
mghavami
0 [mghavami@access1 ~]$ id #other way to show my username
uid=490088225(mghavami) gid=666(clusterusers) groups=666(clusterusers) context=unco
nfin
nfin_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
0 [mghavami@access1 ~]$ uname -a #to show the machine on which i'm connected
Linux access1.aion-cluster.uni.lux 4.18.0-553.34.1.el8_10.x86_64 #1 SMP Mon Dec 16
04:25:43 EST 2024 x86_64 x86_64 x86_64 GNU/Linux
0 [mghavami@access1 ~]$ date #show the date and time
Thu Feb 27 16:03:16 CET 2025
0 [mghavami@access1 ~]$ pwd #show my current working directory
/home/users/mghavami
0 [mghavami@access1 ~]$
```

Figure 1: Answers to the second question with short comments

2 Question 3:

Find the type, and if applicable the file type, of the following commands: `history`, `echo`, `cp`, `srund`, `si`, `python3`, `less`, `zless`, and `aionstat`.

3 Question 4:

What is the command to see your pending jobs? There are two ways (according to my limited knowledge) to answer this question, first by using `sq` as it described in the tip, the second way by using:

`squeue -u USERNAME`

```
mghavami@access1:~  
0 [mghavami@access1 ~]$ type history  
history is a shell builtin  
0 [mghavami@access1 ~]$ type echo  
echo is a shell builtin  
0 [mghavami@access1 ~]$ type cp  
cp is /usr/bin/cp  
0 [mghavami@access1 ~]$ which cp  
/usr/bin/cp  
0 [mghavami@access1 ~]$ file /usr/bin/cp  
/usr/bin/cp: ELF 64-bit LSB shared object, x86-64, version 1 (SYSV), dynamically linked,  
interpreter /lib64/ld-linux-x86-64.so.2, for GNU/Linux 3.2.0, BuildID[sha1]=45ca85136  
599f19562d73893c4d14dc038e3920f, stripped  
0 [mghavami@access1 ~]$ type srun  
-bash: type: srun: not found  
1 [mghavami@access1 ~]$ type srun  
srun is /usr/bin/srun  
0 [mghavami@access1 ~]$ which srun  
/usr/bin/srun  
0 [mghavami@access1 ~]$ file /usr/bin/srun  
/usr/bin/srun: ELF 64-bit LSB executable, x86-64, version 1 (SYSV), dynamically linked,  
interpreter /lib64/ld-linux-x86-64.so.2, for GNU/Linux 3.2.0, BuildID[sha1]=3441af066f  
05a124614f30b8eabb5a6e8f4f418, with debug_info, not stripped, too many notes (256)  
0 [mghavami@access1 ~]$ type si  
si is a function  
  
mghavami@access1:~  
0 [mghavami@access1 ~]$ type si  
si is a function  
si ()  
{  
    local options="";  
    local constraints="batch";  
    while [ -n "$1" ]; do  
        case $1 in  
            -C | --constraint)  
                shift;  
                constraints=${constraints}" "$1  
                ;;  
            *)  
                options=${options}" "$1  
                ;;  
            esac;  
            shift;  
        done;  
        cmd="salloc -p interactive --qos debug -C ${constraints} ${options}";  
        echo "# ${cmd}";  
        $cmd  
    }  
0 [mghavami@access1 ~]$ type python3  
python3 is /usr/bin/python3  
0 [mghavami@access1 ~]$ file /usr/bin/python3  
usr/bin/python3: cannot open 'usr/bin/python3' (No such file or directory)  
0 [mghavami@access1 ~]$ file /usr/bin/python3  
/usr/bin/python3: symbolic link to /etc/alternatives/python3  
0 [mghavami@access1 ~]$ file /etc/alternatives/python3  
/etc/alternatives/python3: cannot open '/etc/alternatives/python3' (No such file or dir  
ectory)  
0 [mghavami@access1 ~]$ which python3  
/usr/bin/python3  
0 [mghavami@access1 ~]$ file /etc/alternatives/python3  
/etc/alternatives/python3: symbolic link to /usr/bin/python3.6  
0 [mghavami@access1 ~]$ file /usr/bin/python3.6  
usr/bin/python3.6: symbolic link to /usr/libexec/platform-python3.6  
0 [mghavami@access1 ~]$ file /usr/libexec/platform-python3.6  
  
mghavami@access1:~  
0 [mghavami@access1 ~]$ which python3  
/usr/bin/python3  
0 [mghavami@access1 ~]$ file /etc/alternatives/python3  
/etc/alternatives/python3: symbolic link to /usr/bin/python3.6  
0 [mghavami@access1 ~]$ file /usr/bin/python3.6  
usr/bin/python3.6: symbolic link to /usr/libexec/platform-python3.6  
0 [mghavami@access1 ~]$ file /usr/libexec/platform-python3.6  
usr/libexec/platform-python3.6: ELF 64-bit LSB shared object, x86-64, version 1 (SYSV)  
, dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, for GNU/Linux 3.2.0, Bu  
ildID[sha1]=a56dbfbb79bdd44556cad022acc610fd7c6a119, stripped  
0 [mghavami@access1 ~]$ type less  
less is /usr/bin/less  
0 [mghavami@access1 ~]$ file /usr/bin/less  
usr/bin/less: ELF 64-bit LSB shared object, x86-64, version 1 (SYSV), dynamically link  
ed, interpreter /lib64/ld-linux-x86-64.so.2, for GNU/Linux 3.2.0, BuildID[sha1]=d689107  
906711e64416d16a936d10b4a100fd73c, stripped  
0 [mghavami@access1 ~]$ type zless  
zless is /usr/bin/zless  
0 [mghavami@access1 ~]$ file /usr/bin/zless  
usr/bin/zless: POSIX shell script, ASCII text executable  
0 [mghavami@access1 ~]$ type aionstat  
aionstat is a function  
aionstat ()  
{  
    sinfo -M aion -h --format=%C | awk -F '/' '{printf "Utilization: %.2f%%\n", $1/$4+1  
00}';  
    printf "\n";  
    pload -M aion -a;  
    printf "\n";  
    qload -M aion -a;  
    printf "\n";  
    echo "Drained nodes: $(sinfo -M aion -h -t drain -o '%D')";  
    printf "%0.5s" {1..50};  
    printf "\n";  
    printf "Jobs status: \n";  
    squeue -M aion -h -o "%t,%r" | sort | uniq -c | sort -r  
}  
0 [mghavami@access1 ~]$
```

Figure 2: Three screenshots of the answers to the third question

4 Question 5:

Once connected to a compute node:

- How can you tell you're now connected to a compute node and not the access node?
First, according to the prompt address, which is different from the access node:
[mghavami@aion-0093 ~](6504253 1N/T/1CN)
then through the `hostname` command.
- What is the name of the compute node you're connected to?
aion-0093
- Use the `sq` to see your job in the queue
- What is your JOBID?
4504293
- What is the total duration of your job?
an hour

```
mghavami@access1:~
0 [mghavami@aion-0093 ~](6504253 1N/T/1CN)$ sq
# squeue -u mghavami
JOBID PARTIT QOS NAME USER NODE CPUS ST TIME TIME_LEFT PRIORITY NODELIST(REASON)
6504253 intera debug interactive mghavami 1 1 R 4:25 55:35 12933 aion-0093
0 [mghavami@aion-0093 ~](6504253 1N/T/1CN)$ hostname
aion-0093
0 [mghavami@aion-0093 ~](6504253 1N/T/1CN)$ lsb_release
LSB Version: :core-4.1-amd64:core-4.1-noarch
0 [mghavami@aion-0093 ~](6504253 1N/T/1CN)$
```

Figure 3: Answers to the fifth question

5 Question 6:

- Collect the execution time of the parallel simulations for 6, 12 and 24 cores.

```
Finalising parallel run
real    3m48.395s
user    0m0.214s
sys     0m0.435s
== Finished job 6504587 at Tue Mar  4 16:06:38 CET 2025
0 [mghavami@aion-0010 motorBike-RUN-06](6504583 1N/T/1CN)$

Finalising parallel run
real    1m49.022s
user    0m0.245s
sys     0m0.402s
== Finished job 6504595 at Tue Mar  4 16:17:20 CET 2025
0 [mghavami@access1 ~]$

Finalising parallel run
real    0m57.794s
user    0m0.207s
sys     0m0.314s
== Finished job 6504596 at Tue Mar  4 16:18:24 CET 2025
0 [mghavami@access1 ~]$

Submitted batch job 6504596
0 [mghavami@aion-0010 motorBike-RUN-12](6504583 1N/T/1CN)$ sq
# squeue -u mghavami
JOBID PARTIT QOS NAME USER NODE CPUS ST TIME TIME_LEFT PRIORITY NODELIST(REASON)
6504595 batch normal motorbike mghavami 1 12 R 1:53 0:07 12119 aion-0010
6504596 batch normal motorbike mghavami 1 24 R 0:00 19:08 12123 aion-0022
6504583 intera debug interactive mghavami 1 6 R 26:46 1:28 12262 aion-0010
0 [mghavami@aion-0010 motorBike-RUN-12](6504583 1N/T/1CN)$
```

Figure 4: Collection of execution time related to question 6

- In your report, create a table with 3 columns: SLURM JOBID, Execution Time and Name of the file log.

SLURM JOBID	Execution Time	Name of the file log
6504587	3m48.395s	SLURM_motorbike_6504587.log
6504595	1m49.022s	SLURM_motorbike_6504595.log
6504596	0m57.794s	SLURM_motorbike_6504596.log

- Copy the three log files in the Git Classroom repository you have cloned. Put them in the logs sub-directory. Don't forget to commit and push.
done

6 Question 7:

Please answer with one line/sentence to each question:

- What Operating System(s) do you use on your personal laptop and/or workstation?
I have Fedora Linux and Windows (Dual boot), but it's been less than a year since I've used Fedora most of the time.
- Are you familiar with Linux and the Command Line? For how long?
I am more or less familiar. I've used Fedora less than a year.
- What programming languages do you know? What level? (beginner, medium, expert)
Matlab, expert
python, medium
- Do you know MPI, OpenMP, CUDA? Or any other parallel programming model?
No
- Why did you choose the Master in HPC?
I'm pursuing the Master in Mathematical Modeling and Computational Science. I've chosen this optional course to learn how to use HPC to be able to solve massive potential (which is highly likely) computational problems.